Late Interaction

ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT
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What the paper proposes

ColBERT delays query-document interaction until after separate encoding and uses MaxSim to preserve fine-grained token evidence.

What changed in this project

Implemented a ColBERT-lite ablation: tokenize the query and dense top-100 passages, embed tokens with the local MiniLM model, compute mean query-token MaxSim, and blend it with the original dense score at 18%.

Measured results

Metric	Baseline	Method	Change
Hit@1	80.0%	85.0%	+5.0 pp
Hit@5	85.0%	90.0%	+5.0 pp
Recall@5	64.7%	68.8%	+4.1 pp
NDCG@10	84.0%	88.1%	+4.2 pp
MRR@20	82.9%	87.5%	+4.6 pp
Latency	26.9 ms	2650.8 ms	—

Inference

This improved ranking quality: Hit@1 rose to 85.0%, Hit@5 to 90.0%, Recall@5 to 68.8%, NDCG@10 to 88.1%, and MRR@20 to 87.5%. The tradeoff is high prototype latency—about 2.65 seconds—because token embeddings are recomputed instead of being persisted in an optimized index.